

A Cyber-Physical System for Freeway Ramp Meter Signal Control Using Deep Reinforcement Learning in a Connected Environment

Yi Hou, Xiangyu Zhang, Peter Graf, Charles Tripp, and David Biagioni

Abstract—Freeway bottlenecks such as on-ramp merging areas account for about 40% of recurring freeway congestion. It is generally agreed that building more roads and adding more lanes to existing infrastructure does not solve the congestion problem, and so dynamic traffic control measures offer a more cost-effective alternative. Ramp meters, traffic signal devices that regulate traffic flow entering freeways, are among the most effective measures to mitigate congestion at on-ramp merging areas on freeways. The confluence of deep reinforcement learning (RL) and connectivity provides a possible solution to advance ramp meter signal control. Deep RL is a group of machine-learning methods that enables an agent learning from the environment to improve its performance. In this study, three deep RL methods—proximal policy optimization (PPO), Ape-X deep Q-network (DQN), and asynchronous advantage actor-critic agents (A3C)—are explored for ramp meter signal control to maximize vehicle speed and traffic throughput, as well as to minimize energy consumption and emissions at freeway on-ramp merging areas in a connected environment. The low computational requirement and scalability of deep RL for deployment make it a powerful optimization tool for time-sensitive applications such as ramp meter signal control. The results of this study show that deep RL methods yield superior performance to both a fixed-time controller and ALINEA, a state-of-the-art feedback controller.

I. INTRODUCTION

Economic and population growth in metropolitan areas will lead to more travel demand and traffic congestions in urban areas. By 2050, about 70% of the world population is projected to live in urban areas [1]. Building more efficient and sustainable mobility systems in urban areas has become the priority of cities and transportation agencies. According to the U.S. Department of Energy [2], bottlenecks such as on-ramp merging areas are responsible for approximately 40% of recurring delay experienced by travelers on freeways. Therefore, improving traffic control on freeway on-ramp merging areas is critical to achieving sustainable mobility systems in urban areas.

Ramp meters have been shown to be effective measures to mitigate congestion at on-ramp merging areas on freeways

and have been widely adopted in urban areas in the United States. Ramp meters are traffic signal devices that regulate the flow of traffic entering a freeway by generating red and green signals using a signal control algorithm. Ramp meter signal control is a complex problem due to the dynamic nature and uncertainty of the transportation system and cannot be described as a mathematical problem with closed-form equations. A number of ramp meter signal algorithms were developed in previous studies. They can be summarized into a few categories including nonparametric feedback control [3], fuzzy control [4], [5], neural network control [6]–[9], model predictive control [10], and hybrid methods [11]. These approaches require large amounts of training data and are usually difficult to replicate at a new location and not generalizable to multiple locations.

In the next decade, vehicles and infrastructure will be more connected and automated than ever before as connected vehicle technology has the potential to improve road safety, reduce travel time, and decrease emissions and fuel consumption [12]. The advancement of deep reinforcement learning (RL) and 5G connectivity provides a unique opportunity to enhance ramp meter signal control performance. Recently, RL has emerged as a broad set of approaches to learn optimal control policies in the face of complex, nonlinear, and stochastic environments. RL has been successfully applied to game playing such as Atari [13] and Go [14], [15], demonstrating an ability to outperform human intelligence in these games, and in many other engineering problems such as in robotics [16], [17], power systems [18], [19], and optimal building operation [20], [21]. A central tenet in RL is that it is possible to iteratively improve upon control policies via direct interaction with the system under study, based on the rewards it receives for taking a specific action in a given system state. RL has also achieved successes in applications of intelligent transportation systems including traffic signal control [22]–[26], autonomous driving [27]–[32], speed limit control [33], [34], and toll road pricing [35], [36]. Recently, a few studies proposed RL for freeway ramp meter signal control. Rezaee [37] developed an RL-based control system that coordinates

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multiple on-ramps in a decentralized structure and showed that RL outperforms the well-known Asservissement Linéaire d'Entrée Autoroutière (ALINEA) [3] algorithm. Fares and Gomaa [38] trained an RL-based density agent to keep the density of the merging area below a critical value and thus maximize freeway utilization. Belletti et al. [39] trained a policy-based RL model adopting Mutual Weight Regularization for ramp meter signal control and achieved control performance similar to ALINEA. Though these existing works have proven effective, they either require prior knowledge to estimate model parameters of traffic dynamics (e.g., fundamental diagram of traffic flow) or historical field data to generate target traffic flow patterns. These works need a lot of human efforts and expert knowledge. Thus, these methods are difficult to be generalized to other new locations.

In this paper, we propose a deep RL training framework for ramp meter signal control to maximize both mobility and energy efficiency in a connected environment where vehicle speeds and positions are acquired constantly in real time within a certain distance from the ramp meter device. Three different deep RL models—proximal policy optimization (PPO), Ape-X deep Q-network (DQN), and asynchronous advantage actor-critic agents (A3C)—are trained and compared with no ramp meter, a fixed-time controller, and ALINEA, a state-of-the-art feedback controller. All deep RL models are trained, validated, and tested in a traffic simulator. The main contributions of this paper are listed below:

- The proposed RL-based ramp meter signal controller is the first of its kind in the context of a connected environment.
- The framework can be generalized to any traffic condition without prior knowledge of traffic parameters or traffic flow pattern as RL models are trained in simulation environments that enable a wide range of scenarios not even observed in historical data. It is ready for deployment given a connected environment.
- All proposed RL algorithms outperform both the fixed-time controller and the state-of-the-art feedback controller, ALINEA.
- Energy and environmental benefits are evaluated and quantified after adopting trained deep RL policies for ramp meter signal control.

The rest of the paper is organized as follows: Section II provides a detailed description of the deep RL methods we proposed, Section III formulates the deep RL training process, Section IV presents the RL results and compare them with baseline approaches, and Section V summarizes the study and discusses conclusions and future research.

II. METHODOLOGY

RL is a branch of machine learning most commonly associated with artificial intelligence for problems that involve sequential decision-making, robotics, and optimal control. In the RL formulation, as shown in Fig. 1, an autonomous agent chooses an action from a set of feasible

actions ($a_t \in \mathcal{A}$) at each step in a control horizon ($t \in \mathcal{T}$) based on its current state ($s_t \in \mathcal{S}$) and a control policy ($\pi : \mathcal{S} \rightarrow \mathcal{A}$). Given the action, the environment returns an immediate reward, r_t . The goal of RL is to obtain the π from its learning experiences that maximizes the expected cumulative discounted reward received from the environment (i.e., $\mathbb{E}_\pi(\sum_{t \in \mathcal{T}} \gamma^t r_t)$, in which $\gamma \in (0,1)$ is the discount factor).

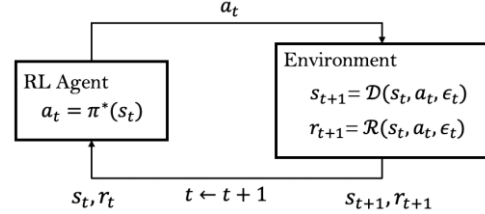


Fig. 1. Illustration of the interaction between RL agent and the environment. $\mathcal{D}$ and $\mathcal{R}$ are the system dynamics and the reward function of the environment, respectively. ϵ_t represents the uncontrollable stochastic disturbance.

Depending on the form of the control policy, π , RL algorithms can be categorized as value-based or policy-based RL. For value-based approaches, value functions, which represent the expected future reward given the current state and action, are first learned from experience. For example, a Q-function approximates the expected discounted reward that would be received by taking action a while in state s and then following policy π , and is defined as [40]:

$$Q_\pi(s, a) = \mathbb{E}_\pi \left[\sum_{\tau=t}^{|\mathcal{T}|} \gamma^{\tau-t} r_\tau \mid s_t = s, a_t = a \right]. \quad (1)$$

With $Q_\pi(s, a)$ learned using approaches such as Monte Carlo learning, SARSA (state-action-reward-state-action) learning [41], and Q-learning, a greedy policy

$$a_t = \pi(s_t) = \underset{a \in \mathcal{A}}{\operatorname{argmax}} Q(s_t, a) \quad (2)$$

is obtained. Algorithms of this type are called value-based RL and are only implicitly defined via the maximization problem (2).

In contrast, another type of policy does not depend on value function; instead, they are direct parameterized functions of the state (i.e., $\pi_\theta(s_t)$, in which θ is the parameter vector). As a result, the control performance, which depends on π_θ , becomes a function of θ . To learn the optimal policy, a policy-based algorithm can apply stochastic gradient ascent at each learning iteration, as shown in (1), until convergence to the optimal parameter θ^* is obtained. The gradient term $\hat{\nabla}_\theta J(\theta)$ is estimated from experience collected in the current learning iteration

$$\theta^{t+1} = \theta^t + \alpha \hat{\nabla}_\theta J(\theta) = \theta^t + \alpha \hat{\nabla}_{\mathbb{E}_{\pi_\theta}} \left[\sum_{t \in \mathcal{T}} \gamma^t r_t \right]. \quad (3)$$

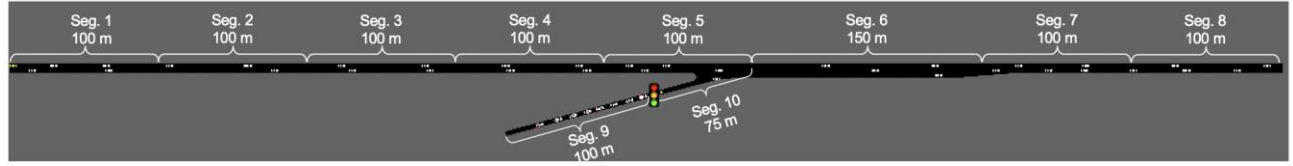


Fig. 2. An on-ramp merging area on a two-lane freeway in a SUMO simulator.

A group of hybrid methods are actor-critic algorithms, which learn both policy (actor) and value functions (critic). Compared with pure policy-based algorithms, actor-critic approaches can use the critic to reduce variance when estimating $\nabla_{\theta} J(\theta)$ [42].

In this study, we explore both value-based and actor-critic approaches. We choose two actor-critic algorithms, PPO [43] and A3C [44], and one value-based algorithm, Ape-X DQN [45], due to their popularity and successful implementations demonstrated in the past decade.

III. EXPERIMENTS

A. Experiment Setup

The objective of the study is to train deep RL ramp meter signal controllers to maximize both mobility and energy efficiency at the freeway on-ramp merging area. We ran RL experiments using Flow [46], an open-source RL framework that provides an interface between the microscopic traffic simulator Simulation of Urban MObility (SUMO) [47] and a popular RL library, RLlib [48]. A typical two-lane freeway on-ramp merging area with a ramp meter device deployed on a one-lane ramp at the entrance of the freeway was set up in Flow, as depicted in Fig. 2. In the experiments, we simulated a connected environment where vehicle speeds and positions are acquired constantly in real time within a certain distance from the ramp meter device. The speed limit on the freeway was set as 65 mph, whereas the speed limit on the ramp was set as 40 mph. With the default car following model parameters in SUMO, the observed bottleneck capacity (or maximum throughput) of the merging area bottleneck in simulations is between 3,600 and 3,700 vehicles per hour (vph). For each simulation episode, we ran 1,000 time steps with each time step as one second. Before each simulation starts, we warmed up simulation with 40 time steps to load traffic on the simulation network. In order to train deep RL controllers that are capable of optimizing traffic in conditions where traffic inflow is both below and above bottleneck capacity, the simulator generated traffic inflow varying every 100 time steps on both the freeway and ramp in each simulation episode. Because ramp meters are deployed mainly for congestion mitigation at bottlenecks, RL policies should be trained with congested or near-congested traffic data. A traffic inflow variation with an expected value slightly higher than bottleneck capacity is appropriate. Thus, more specifically, in every 100 time steps, the traffic inflow on the freeway was reset following a normal distribution with mean of 3,000 vph and standard deviation of 300 vph and with value clipped between 0 and 3,900 vph, while the traffic inflow on the ramp was reset following a normal distribution with mean

of 900 vph and standard deviation of 300 vph and with value clipped between 0 and 1,500 vph.

B. Action and State Space

Because ramp meters regulate traffic flow by generating red or green signals to stop vehicles or let them enter the freeway, the action at each time step is binary: red or green. In order to reduce the complexity of the state space, instead of using speeds and positions of all vehicles on the road network, we aggregated the raw vehicle data at a segment level. As shown in Fig. 2, the freeway and on-ramp were divided into many small segments. All segments were 100 meters long except segment 6 at the merging area with a length of 150 meters and segment 10 at the entrance of the freeway with a length of 75 meters. Therefore, the state space has 40 dimensions and comprises the following three components:

1. Mean speed and number of vehicles on each lane of each segment, which contains a total of 38 variables ($2 \text{ lanes} \times 2 \text{ variables} \times 7 \text{ segments} + 1 \text{ lane} \times 2 \text{ variables} \times 2 \text{ segments} + 3 \text{ lanes} \times 2 \text{ variables} \times 1 \text{ segment} = 38$).
2. Ramp meter signal phase (green or red) of the previous simulation step.
3. Traffic throughput at the freeway merging area bottleneck over the last 30 seconds.

C. Reward Function

The reward function design is critical to the success of RL training. The reward function together with the action space define the optimal policy, which will choose the action that maximizes the expected discounted reward at every time step. For the purpose of maximizing mobility and energy efficiency and mitigating congestion at the freeway on-ramp merging area, the reward function was designed to penalize congestions or queues developed at the merging area (segment 6) as well as upstream of the ramp meter device (segment 9) while rewarding fast vehicle movement on both the merging area and the ramp. Therefore, the reward function is defined as:

$$R(s_t, a_t) = \frac{\bar{v}_{6,9,10}}{s_{max}} + c_m + c_r \quad (4)$$

where $\bar{v}_{6,9,10}$ is denoted as mean speed of vehicles on segment 6, 9, and 10, s_{max} is speed limit on the freeway to normalize speed reward, c_m is a congestion penalty at the merging area (segment 6), and c_r is a penalty for the queue spilling back on the on-ramp (segments 9 and 10). c_m equals -1 if 5 or more vehicles on segment 6 have speed lower than 1 m/s, 0 otherwise. Empirical observations in simulations showed that

once 5 or more vehicles are stopped in traffic, usually the congestion will persist unless action is taken. c_r equals -1 if 9 or more vehicles on segment 9 have speed lower than 1 m/s, 0 otherwise. The rationale behind this is that the ramp segment upstream of the ramp meter device can only hold a maximum of 9 vehicles. A trial-and-error process of tuning the values of c_m and c_r showed that the above-mentioned values of c_m and c_r yielded the best model performance. The discount rate of the reward function is set as 0.99 , which is the default value in RLlib.

IV. EXPERIMENT RESULTS AND ANALYSIS

We ran RL experiments with all three algorithms—PPO, Ape-X DQN, and A3C—with a horizon of $1,000$ time steps using 32 CPUs. Each algorithm was refined by tuning its hyperparameters that have the most impacts on training results, adopting the trial-and-error strategy. PPO was trained with a sample size of $32,000$ time steps per training iteration and a learning rate of $5e-5$. Ape-X DQN was trained with a sample size of $1,000$ time steps per iteration, an initial exploration fraction of 0.1 , a final exploration fraction of $1e-4$, a learning rate of $5e-4$, and a replay buffer size of 5 million (M) time steps. A3C was trained with a sample size of 500 per training iteration and a learning rate of $1e-4$ on each worker. All RL policies trained using the above-mentioned algorithms were implemented using a neural network with one hidden layer of 128 neurons and a tanh activation function. Fig. 3 presents the reward curves of all three algorithms over the training cycle. The reward curves plateaued at the end of training, meaning that all trainings have converged. PPO, Ape-X DQN, and A3C converged with rewards around 600 , 500 , and 550 , respectively. The reward curves of PPO and Ape-X DQN flatten after about $20M$ training steps (5 and 4 wall clock hours respectively), whereas it took about $300M$ training steps (42 wall clock hours) for the reward curve of A3C to flatten.

The results of deep RL algorithms were benchmarked against baseline scenarios including the feedback controller ALINEA, no ramp meter, and a fixed-time controller. The details of the baseline scenarios are described below:

- **No ramp meter:** No ramp meter is installed at the entrance of the freeway. If the freeway traffic allows, vehicles enter the freeway immediately when they reach the end of the on-ramp.
- **Fixed-time controller:** A ramp meter is installed at the freeway entrance with fixed red and green phases. In the process of manually tuning the fixed timing plan, a total cycle time of 4 seconds consisting of a fixed 3 -second red phase and a fixed 1 -second green phase yields the best performance and was selected for model comparison. One vehicle per lane is allowed to enter the freeway during the green phase.
- **ALINEA:** A ramp meter with an ALINEA controller is installed at the entrance of the freeway. The cycle time of the ALINEA controller in this baseline consists of a 1 -second green phase and a variable red phase.

One vehicle is allowed to enter the freeway in each lane during the green phase. The cycle time c is estimated using

$$c = \frac{N}{q_{desire} - q_{freeway}} \cdot 3600, \quad (5)$$

where N is the number of lanes on the ramp, q_{desire} is the desired total traffic inflow at the merging area bottleneck in vph, and $q_{freeway}$ is the traffic inflow on the freeway. q_{desire} at time step t is determined by

$$q_{desire}(t) = q_{total}(t-1) + C_f(n^* - n_{t-1}), \quad (6)$$

where $q_{total}(t-1)$ is the total traffic inflow from both freeway and ramp at time step $t-1$, C_f is a feedback coefficient, n^* is the critical number of vehicles at the merging area that yield the maximum traffic outflow, and n_{t-1} is the number of vehicles at the merging area at time step $t-1$. In this case, the cycle time is updated every 15 second. We use $n^* = 10$ and $C_f = 20$, as these values yield the best controller performance.

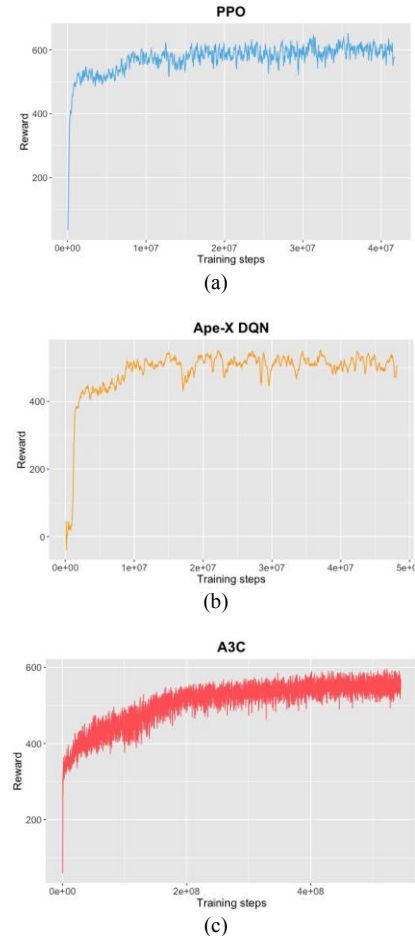


Fig. 3. Reward curves of (a) PPO, (b) Ape-X DQN, and (c) A3C.

The trained deep RL ramp meter signal controllers were first evaluated with the same traffic inflow setting during RL training, which generates mixed near-congested and congested traffic conditions. Because the primary purpose of a ramp meter is to mitigate traffic congestions at the freeway merging area bottleneck, the trained deep RL ramp meter signal controllers were then further tested in extremely congested traffic conditions where traffic inflow is 8% to 50% higher than the bottleneck capacity. In the extremely congested test case, freeway inflow was set as constant 3,900

vph and the on-ramp inflow was set the same as during RL training, varying between 0 and 1,500 vph every 100 time steps following a normal distribution. The results were analyzed from perspectives of both mobility and energy. The final results of simulations included the following metrics:

- Traffic throughput (vph)
- Average vehicle speed (m/s)
- Average number of stops per vehicle
- Average vehicle fuel efficiency (mpg)

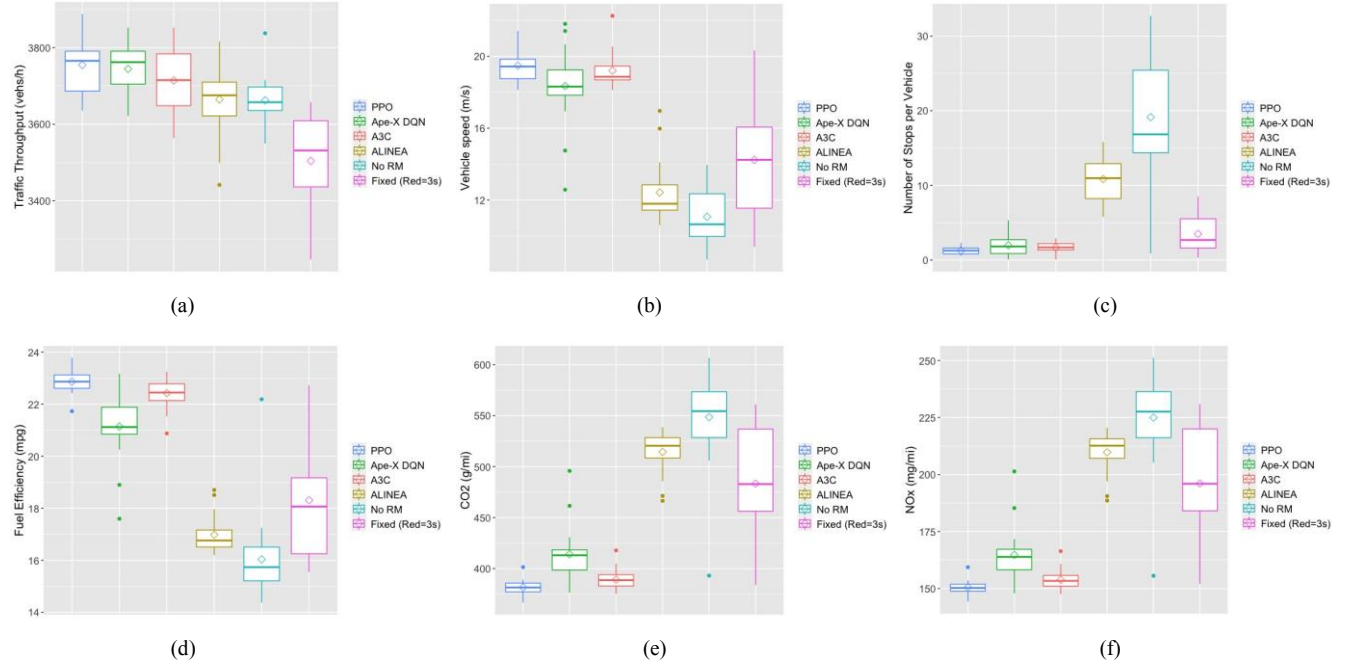


Fig. 4. Comparison of deep RL algorithms and baseline performance in mixed near-congested and congested traffic conditions for (a) traffic throughput, (b) vehicle speed, (c) number of stops per vehicle, (d) fuel efficiency, (e) CO₂ emissions, and (f) NO_x emissions.

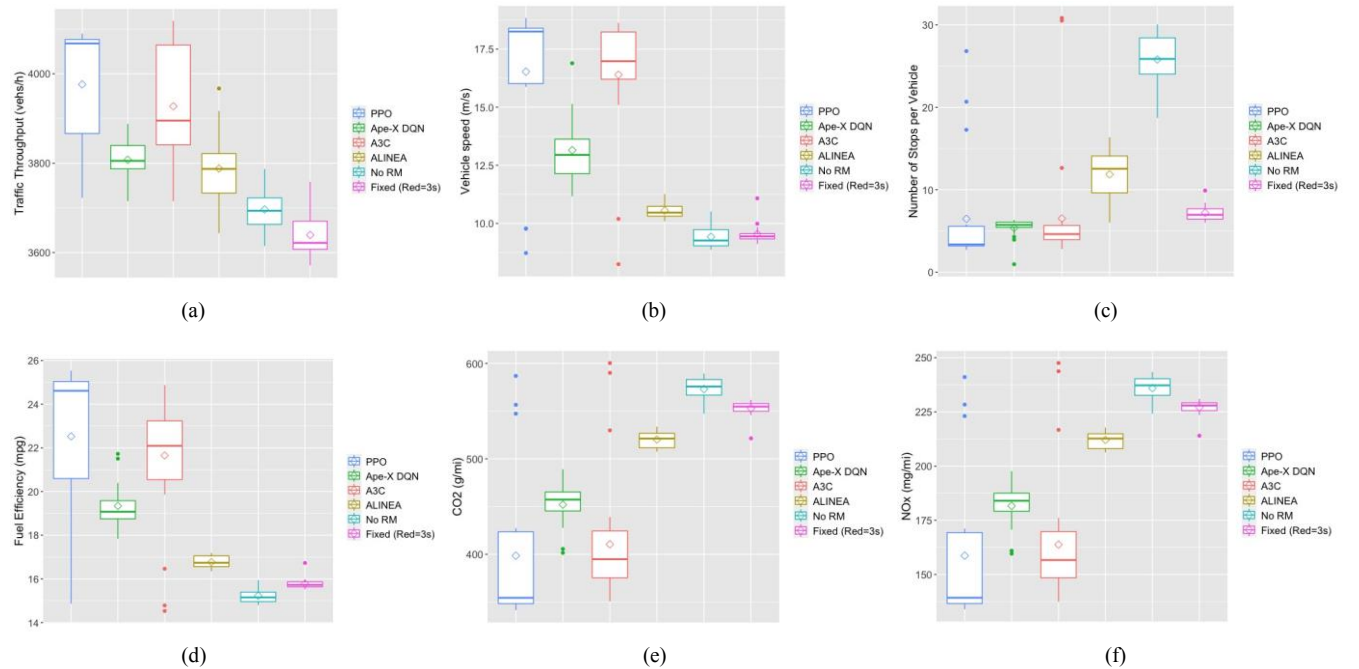


Fig. 5. Comparison of deep RL algorithms and baseline performance in extremely congested traffic conditions for (a) traffic throughput, (b) vehicle speed, (c) number of stops per vehicle, (d) fuel efficiency, (e) CO₂ emissions, and (f) NO_x emissions.

- Average greenhouse gas (GHG) emission per vehicle including CO₂ (g/mi) and NO_x (mg/mi).

The energy and emission model used in this study is the default emission model in SUMO, namely HBEFA3/PC_G_EU4 [49]. In order to achieve statistical significance for the performance results, we ran 20 simulation episodes for each trained deep RL policy and baseline scenario. For each deep RL policy and baseline scenario, the results were averaged across the 20 simulation runs.

The comparison of results between deep RL policies and baselines in mixed near-congested and congested traffic conditions are displayed in Fig. 4. Boxplots are used to show the distribution of the performance metrics and the diamond shape is used to show the mean value. As shown in Fig. 4, deep RL algorithms yielded both higher mean and median values in traffic throughput, vehicle speed, and fuel efficiency than the baselines, and lower mean and median values in number of stops and GHG emissions, indicating that deep RL policies outperformed all the baseline scenarios. As shown in Table I, when compared with ALINEA, no ramp meter, and the fixed-time controller, all three deep RL policies, on average, respectively increased traffic throughput by 2%, 2%, and 7%; increased vehicle speed by 53%, 72%, and 34%; reduced number of stops per vehicle by 85%, 91%, and 53%; increased fuel efficiency by 30%, 38%, and 21%; reduced CO₂ emissions by 23%, 28%, and 18%; and reduced NO_x emissions by 25%, 31%, and 20%. Optimizing the reward by penalizing congestion and rewarding traffic speed at the merging area and on-ramp allowed for avoiding congestion and reducing speed variance at the bottleneck, and therefore led to higher traffic throughput, lower number of vehicle stops, higher energy efficiency, and lower emissions. Of all three deep RL policies, PPO produced the best performance in each category of comparison.

The same performance metrics were examined in extremely congested traffic conditions where traffic inflow is 8% to 50% higher than the bottleneck capacity. The results

TABLE I
PERFORMANCE METRIC CHANGES WHEN COMPARED WITH BASELINE IN MIXED NEAR-CONGESTED AND CONGESTED TRAFFIC CONDITIONS

Performance Metric	RL Average	Changes Compared with Baseline		
		ALINEA	No Ramp Meter	Fixed
Traffic throughput (vph)	3,738	+2%	+2%	+7%
Average vehicle speed (m/s)	19.0	+53%	+72%	+34%
Average number of stops per vehicle	1.7	-85%	-91%	-53%
Average vehicle fuel efficiency (mpg)	22.1	+30%	+38%	+21%
Average CO ₂ emissions per vehicle (g/mi)	395	-23%	-28%	-18%
Average NO _x emissions per vehicle (mg/mi)	156	-25%	-31%	-20%

TABLE II
PERFORMANCE METRIC CHANGES WHEN COMPARED WITH BASELINE IN EXTREMELY CONGESTED TRAFFIC CONDITIONS

Performance Metric	RL Average	Changes Compared with Baseline		
		ALINEA	No Ramp Meter	Fixed
Traffic throughput (vph)	3,904	+3%	+6%	+7%
Average vehicle speed (m/s)	15.4	+45%	+63%	+61%
Average number of stops per vehicle	6.1	-48%	-76%	-15%
Average vehicle fuel efficiency (mpg)	21.2	+26%	+39%	+34%
Average CO ₂ emissions per vehicle (g/mi)	420	-19%	-27%	-24%
Average NO _x emissions per vehicle (mg/mi)	168	-21%	-29%	-26%

are presented in Fig. 5. Deep RL policies demonstrated even larger advantages when compared with baselines in terms of traffic throughput. As shown in Table II, the trained deep RL policies had, on average, 3%, 6%, and 7% more traffic throughput than ALINEA, no ramp meter signal control, and fixed-time control, respectively, in extremely congested traffic conditions, as opposed to 2%, 2%, and 7% higher in mixed near-congested and congested traffic conditions. For other performance metrics, all three trained deep RL policies had, on average, 45%, 63%, and 61% higher speed; 48%, 76%, and 15% fewer number of stops per vehicle; 26%, 39%, and 34% higher fuel efficiency; 19%, 27%, and 24% less CO₂ emissions; and 21%, 29%, and 26% less NO_x emissions than ALINEA, no ramp meter, and fixed-time control, respectively. Comparing among deep RL policies, PPO outperformed Ape-X DQN and A3C for all performance metrics except the number of stops per vehicle, which was lowest for Ape-X DQN.

Vehicle speeds at different locations along the freeway were further analyzed and compared between deep RL ramp meter controllers and the baselines in mixed near-congested

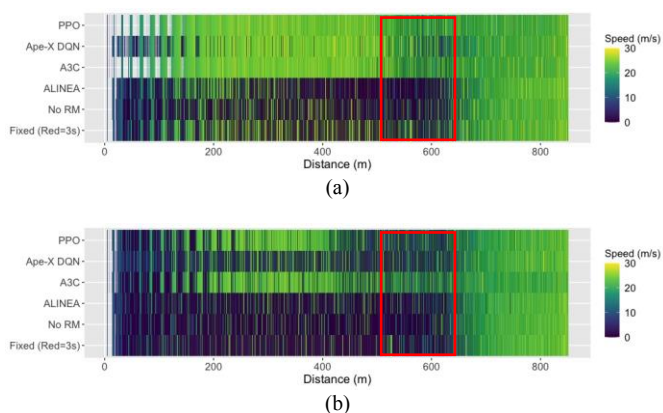


Fig. 6. Vehicle speed at all longitude locations along the freeway for (a) mixed near-congested and congested traffic conditions and (b) extremely congested traffic conditions.

and congested traffic conditions and extremely congested traffic conditions. Fig. 6 displays the vehicle speed at all longitude locations for all deep RL controllers and baselines. The x-axis is the freeway longitude coordinates. Lighter colors represent higher speeds, whereas darker colors represent slower speeds. The red rectangles mark the merging area. In Fig. 6(a), deep RL ramp meter controllers produced higher vehicle speeds and less speed variance upstream and at the merging area than the baselines in mixed near-congested and congested traffic conditions. The merging area with deep RL controllers shows vehicle slowdown, whereas the merging area in baseline scenarios is completely congested. Fig. 6(b) shows that congestions developed in both deep RL controller and baseline scenarios in extremely congested traffic conditions where traffic inflow is 8% to 50% higher than the bottleneck capacity. However, the speed comparison upstream and at the merging area demonstrates that deep RL controllers are still effective in smoothing traffic and mitigating traffic congestions even in extremely congested traffic conditions.

V. CONCLUSIONS AND FUTURE WORK

In this study, three deep RL algorithms—PPO, Ape-X DQN, and A3C—were used to train control policies for ramp meter signal control at a freeway on-ramp in the SUMO simulator using Flow and RLlib. The results show that the trained deep RL policies outperform the baselines—feedback controller ALINEA, no ramp meter, and fixed-time controller—in terms of traffic throughput, vehicle speed, number of stops per vehicle, vehicle fuel efficiency, and GHG emissions including CO₂ and NO_x in mixed near-congested and congested traffic conditions and extremely congested traffic conditions. Of all three deep RL algorithms, PPO slightly outperformed the other two in all evaluation metrics and converged at a similar rate as Ape-X DQN but about 15 times faster than A3C.

The use of deep RL for ramp meter signal control is appealing because it does not require formulating and solving an explicit optimization model and does not need prior knowledge to estimate model parameters. Instead, good solutions are found by repeatedly evaluating RL-based decisions via forward simulation and then using reward feedback from the simulation to incrementally improve the policy. Compared with most classical control methods, RL is highly robust to dynamic, stochastic problems such as ramp meter signal control. Instead of reoptimizing from scratch every time traffic conditions change, RL can sidestep this issue by training a policy in the context of stochastic scenarios that reflect the expected, dynamic behavior of the real-world system. In this case, RL can be simulation-trained, where the RL agent is trained on many simulations scenarios of the system under consideration to try different actions and learn the optimal policy. For practical applications, an RL agent might be trained with a simulator and then be allowed to further refine its policy in the real world. This would also allow it to adapt to changing traffic patterns.

This study only evaluated the effectiveness of deep RL in the control of a single ramp meter device in a simulator. In future research work, we will translate a trained RL policy in a simulator to real-world systems and develop a method to systematically tune the reward parameters for arbitrary ramps. Future research efforts will also focus on exploring coordination of multiple ramp meters on freeway networks to optimize freeway traffic at a systematic level using multi-agent RL. Other RL optimization strategies such as evolution strategies will also be explored in the future to improve ramp meter signal control performance.

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